

ASSESSMENT TOOL-3

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COURSE CODE: CSA1511

COURSE NAME: Cloud Computing and Big Data Analysis

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud computing is a model for delivering computing resources such as servers, storage, databases, networking, software, and processing power through the Internet on demand. Users can access these resources without owning or maintaining the physical infrastructure, usually using a pay-as-you-go pricing model.

Four Key Characteristics

1. On-Demand Self-Service

Users can automatically provision computing resources such as virtual machines and storage whenever required, without direct interaction with the service provider.

2. Broad Network Access

Cloud services are accessible through the Internet using different devices such as laptops, smartphones, tablets, and desktop computers.

3. Resource Pooling

Cloud providers use shared physical resources to serve multiple customers. Virtualization allows computing resources to be dynamically allocated based on demand.

4. Rapid Elasticity

Cloud resources can be quickly scaled up or down according to workload. For example, an online shopping application can increase server capacity during a festival sale and reduce it afterward.

Difference from Traditional On-Premise Computing

Cloud Computing	Traditional On-Premise Computing
Resources are accessed over the Internet	Resources are owned and operated by the organization
Pay-as-you-use model	High upfront hardware investment
Easily scalable	Scaling requires purchasing new hardware
Provider manages physical infrastructure	Organization manages infrastructure

Example: A company using a cloud platform can launch additional virtual servers during high traffic without purchasing physical servers.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Cloud computing evolved through several important technological developments:

Mainframe Computing → Personal Computers → Client-Server Computing → Internet → Virtualization → Web Services → Modern Cloud Computing

1. Mainframe Computing

In the early stage, powerful mainframe computers provided centralized processing to many users through terminals. This introduced the idea of sharing computing resources.

2. Personal Computers

The development of PCs moved computing power from centralized systems to individual users. However, organizations still needed servers for centralized data and applications.

3. Client-Server Computing

Applications were divided between client devices and centralized servers. This improved resource sharing and network-based application access.

4. Internet Evolution

The growth of the Internet enabled users to access applications and data from different geographical locations. This created the foundation for remote computing services.

5. Server Virtualization

Virtualization allowed a single physical server to run multiple **virtual machines**. This improved hardware utilization, reduced costs, and enabled flexible resource allocation.

6. Web Services and APIs

Web services and APIs allowed software applications to communicate over networks. Computing capabilities could now be delivered as reusable services.

7. Modern Cloud Platforms

Cloud providers combined **large-scale data centers, virtualization, high-speed networks, automation, APIs, and distributed computing** to create modern cloud platforms.

How These Changes Enabled Cloud Computing

The evolution of hardware increased processing capacity, virtualization improved resource utilization, the Internet enabled global access, and web services enabled software-based service delivery. Together, these technologies made possible the **scalable, flexible, on-demand cloud platforms** used today.

Example: Modern cloud providers can dynamically allocate virtual servers and storage to users through web portals or APIs.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Server virtualization is a technology that divides a single physical server into multiple virtual machines (VMs) using a software layer called a hypervisor. Each VM can run its own operating system and applications independently.

Role in Cloud Computing

- **Resource Utilization:** Maximizes the use of physical servers.
- **Resource Sharing:** Allows multiple users or applications to share hardware securely.
- **Scalability:** Enables virtual resources to be created or removed quickly.
- **Cost Reduction:** Reduces the need for large amounts of physical hardware.
- **Isolation:** Keeps applications and workloads separated from each other.
- **Fast Deployment:** New virtual machines can be created within minutes.

Virtualization vs Cloud Computing

Virtualization	Cloud Computing
An enabling technology	A service delivery model
Creates virtual machines from physical hardware	Delivers computing resources as services
Uses hypervisors to manage VMs	Uses virtualization, networking, automation, security, and management
Can operate in private data centers	Provides on-demand services over networks
Example: VMware or Hyper-V VM	Example: Amazon EC2

Key Difference

Virtualization is the technology that abstracts physical resources, whereas cloud computing uses virtualization along with other technologies to deliver flexible, scalable, and on-demand services.

Example: A cloud provider may use virtualization to create thousands of virtual servers and offer them to customers as an **IaaS service**.

4. Discuss the role of data centers in delivering cloud services..

A data center is a specialized facility containing physical servers, storage systems, networking equipment, power supplies, cooling systems, and security infrastructure. It forms the physical foundation of cloud computing.

Major Roles

Compute Resources

Physical servers provide processing power for running cloud applications and virtual machines.

- 1. Data Storage**

Storage systems maintain user files, databases, application data, and backups.

- 2. Networking**

High-speed networks connect servers and allow users to access cloud services through the Internet.

- 3. Virtualization**

Data centers use virtualization to divide physical resources into multiple virtual machines and allocate them efficiently.

- 4. Security**

Physical and digital security mechanisms protect cloud infrastructure and customer data.

- 5. High Availability**

Backup power, redundant servers, and multiple network connections help maintain continuous service availability.

Example

When a user accesses a cloud application, the request travels through the Internet to the provider's data center. The data center processes the request using servers and databases and sends the result back to the user.

Therefore, data centers provide the physical infrastructure, while cloud technologies convert these resources into flexible services accessible over the Internet.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Model	Full Form	What It Provides	User Responsibility	Real-World Example
IaaS	Infrastructure as a Service	Virtual machines, storage, networking	Manages OS, applications, and data	Amazon EC2
PaaS	Platform as a Service	Development tools, runtime, middleware	Develops and manages applications	Google App Engine
SaaS	Software as a Service	Ready-to-use software	Mainly uses and configures the software	Google Workspace
XaaS	Everything as a Service	Various IT resources delivered as services	Depends on the specific service	AWS Lambda (FaaS)

IaaS – Infrastructure as a Service

IaaS provides basic computing infrastructure such as virtual servers, storage, and networks. The customer has greater control but must manage the operating system and applications.

Example: Amazon EC2 provides virtual servers that organizations can configure according to their requirements.

PaaS – Platform as a Service

PaaS provides a ready-made development and deployment environment. Developers can build and deploy applications without managing the underlying

Example: Google App Engine servers and operating systems allows developers to deploy applications while the cloud provider manages the infrastructure.

SaaS – Software as a Service

SaaS provides complete software applications over the Internet. Users access the application through a browser or app, while the provider manages the infrastructure, platform, updates, and maintenance.

Example: Google Workspace provides cloud-based applications such as Gmail, Docs, and Drive.

XaaS – Everything as a Service

XaaS is a broad concept where different IT capabilities are delivered as cloud-based services. Examples include Database as a Service (DBaaS), Storage as a Service (STaaS), Security as a Service (SECaaS), and Function as a Service (FaaS).

Example: AWS Lambda provides Function as a Service, allowing developers to run code without managing servers.

Conclusion:

The main difference is the level of management and control given to the user. IaaS provides the most control, PaaS reduces infrastructure management, SaaS provides ready-to-use software, and XaaS represents the broader concept of delivering IT capabilities as cloud services.